

Sulyab Thottungal Valapu

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Research Statement

My research uses **Internet measurement** to understand the **real-world effects of the Internet's constant evolution**. This evolution may be driven by new protocols and standards (such as the shift from IPv4 to IPv6), changes in deployment practices (such as the rise of cloud computing), or the emergence of new applications (such as “Web3”). I focus on **developing methods to measure how these changes are adopted in practice** and on **understanding the unexpected interactions between “old” and “new” mechanisms**, with implications for **usability, security, and privacy**. Recently, I have been studying the subtle risks of DNS coexisting with alternative naming systems, as well as IPv6 adoption and the privacy risks and opportunities created by the shift to IPv6. Previously, I worked on designing decentralized reputation mechanisms for blockchain applications.

Experience

- 2021 – ♦ **University of Southern California** | Graduate Teaching Assistant
- Teaching Assistant for *Fundamentals of Computation* (Spring '21, Fall '24), *Intro to Algorithms* (Fall '21), *Intro to Computer Systems* (Spring '25, Fall '25), *Intro to Computer and Network Security* (Spring '22), and *Database Systems* (Summer '24, Summer '25).
- 2018 – 2020 ♦ **Arista Networks** | Software Engineer
- Developed Network Layer software for Arista EOS®
 - Involved in all stages of software development process including design, development, deployment, debugging and maintenance

Education

- 2021 – ♦ **Ph.D., University of Southern California** Computer Science.
- Advised by [Prof. John Heidemann](#)
 - Member of [Analysis of Network Traffic \(ANT\) Lab](#)
 - GPA: 3.96/4.0
- 2014 – 2018 ♦ **B.Tech., IIIT Allahabad** Information Technology.
- GPA: 9.37/10. Graduated with Honors.

Publications

Conference Proceedings

- 1 **Sulyab Thottungal Valapu** and John Heidemann. “Towards a Non-Binary View of IPv6 Adoption”. In: *Proceedings of the 2025 ACM Internet Measurement Conference*. IMC '25. New York, NY, USA: Association for Computing Machinery, Nov. 21, 2025, pp. 727–745. ISBN: 979-8-4007-1860-1. DOI: [10.1145/3730567.3764467](https://doi.org/10.1145/3730567.3764467).
- 2 Alex Sotiropoulos, **Sulyab Thottungal Valapu**, Linus Lei, Jared Coleman, and Bhaskar Krishnamachari. “Poster: Crowd-SFT: Crowdsourcing for LLM Alignment”. In: *2025 IEEE International Conference on Decentralized Applications and Infrastructures (DAPPS)*. ISSN: 2835-3498. July 2025, pp. 135–136. DOI: [10.1109/DAPPS65174.2025.00026](https://doi.org/10.1109/DAPPS65174.2025.00026).

- 3 Dipsy Desai, Kicho Yu, and **Sulyab Thottungal Valapu**. “Geofeed Adoption and Authentication”. In: *2025 IEEE Network Operations and Management Symposium (NOMS)*. ISSN: 2374-9709. May 2025, pp. 01–08.  DOI: [10.1109/NOMS57970.2025.11073659](https://doi.org/10.1109/NOMS57970.2025.11073659).
- 4 **Sulyab Thottungal Valapu**, Aritri Saha, Bhaskar Krishnamachari, Vivek Menon, and Ujjwal Guin. “Reward-Based Blockchain Infrastructure for 3D IC Supply Chain Provenance”. In: *2025 IEEE International Symposium on Hardware Oriented Security and Trust (HOST)*. ISSN: 2765-8406. May 2025, pp. 370–380.  DOI: [10.1109/HOST64725.2025.11050057](https://doi.org/10.1109/HOST64725.2025.11050057).
- 5 **Sulyab Thottungal Valapu**, Tamoghna Sarkar, Jared Coleman, Anusha Avyukt, Hugo Embrechts, Dimitri Torfs, Michele Minelli, and Bhaskar Krishnamachari. “DARSAN: A Decentralized Review System Suitable for NFT Marketplaces”. In: *Blockchain – ICBC 2023*. Ed. by Qin Wang, Jun Feng, and Liang-Jie Zhang. Cham: Springer Nature Switzerland, 2023, pp. 3–20. ISBN: 978-3-031-44920-8.  DOI: [10.1007/978-3-031-44920-8_1](https://doi.org/10.1007/978-3-031-44920-8_1).

Preprints

- 1 Alex Sotiropoulos, **Sulyab Thottungal Valapu**, Linus Lei, Jared Coleman, and Bhaskar Krishnamachari. *Crowd-SFT: Crowdsourcing for LLM Alignment*. June 2025.  DOI: [10.48550/arXiv.2506.04063](https://doi.org/10.48550/arXiv.2506.04063). arXiv: [2506.04063\[cs\]](https://arxiv.org/abs/2506.04063).

Technical Reports

- 1 **Sulyab Thottungal Valapu** and Bhaskar Krishnamachari. *A Survey of Probabilistic Micropayment Schemes*. Tech. rep. IEEE Blockchain Technical Briefs, Mar. 2022.

Skills

Languages

Proficient	 Python • C • C++
Familiar	 Rust • WebAssembly • Java

Tools & Technologies

Systems & Networking	 Linux kernel networking stack (incl. netfilter) • tcpdump / Wireshark • Linux system programming • OpenWrt
Virtualization	 Containers (incl. Docker) • Virtual Machines (VMs)
Data Processing & Analysis	 SQL • DuckDB (incl. Ducklake) • NumPy • Pandas • Apache Arrow • Jupyter
Developer Tools	 Git • Shell scripting (Bash)
Other	 $\LaTeX$

Miscellaneous Experience

Internships

- 2017/05 - 2017/07  **Yonsei University, South Korea**
- Designed and implemented a blockchain-based protocol for intelligent autonomous vehicles to exchange messages and purchase services.
 - Explored the concept of blockchain branching to tackle scaling limitations

Miscellaneous Experience (continued)

Awards and Achievements

2017 ◇ Dean's Merit List, IIT Allahabad

Activities

2017 ◇ Core Team member of **Hack In The North**, the largest student-led hackathon in India
2022, 2023 ◇ Research Coach for the **Viterbi Summer Institute**, a program by the USC Viterbi School of Engineering designed to offer incoming freshmen from underrepresented backgrounds an introductory research experience.

Selected Coursework

Systems	◇ Advanced Computer Networking (<i>Graduate level</i>) Networked Systems in Cloud Computing (<i>Graduate level</i>) Internet Measurement (<i>Graduate level</i>) Autonomous Cyber-Physical Systems (<i>Graduate level</i>) Applied Cryptography (<i>Graduate level</i>) Network Security Operating Systems
Algorithms	◇ Advanced Analysis of Algorithms (<i>Graduate level</i>)
AI	◇ Cognitive Science Natural Language Processing Artificial Intelligence